

ICLR2027

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Paper under double-blind review

ABSTRACT

Distilling the advanced reasoning capabilities of large language models (LLMs) into lightweight student models remains a critical challenge. Existing offline distillation forces students to imitate overly complex teacher trajectories, yielding inefficient knowledge transfer. Meanwhile, online distillation, though enabling autonomous exploration, suffers from error accumulation and severe instability in long-horizon reasoning tasks. To address these limitations, we propose a Capability-Aware Reasoning Distillation framework coupling offline data adaptation with online self-correction. Off-policy, we introduce a student-perplexity-aware data curation pipeline to filter low-information redundancies and construct a progressive reasoning curriculum. On-policy, we design a Teacher-Anchored Local Alignment mechanism. By monitoring step-level cognitive states, the system triggers a lossless Key-Value (KV) cache rollback upon detecting logical deviations, forcing the student to re-explore viable paths within its capability boundary. Optimized via an RL-free Hierarchical Preference Alignment loss, our framework reinforces autonomously discovered correct logic while applying soft teacher regularization to smoothly elevate the reasoning ceiling. Extensive experiments across mathematical, medical, and agent-based reasoning benchmarks demonstrate that our lightweight models significantly surpass conventional offline and online distillation baselines in both generalization and reasoning stability.

1 INTRODUCTION

Large language models (LLMs) have recently achieved remarkable progress in complex reasoning tasks, including mathematical problem solving, scientific reasoning, and agent-based decision making. As reasoning models continue to scale in both model size and training cost, transferring reasoning capabilities from strong teacher models to smaller and more efficient students has become an increasingly important research direction. Among existing approaches, chain-of-thought (CoT) distillation has emerged as a practical and scalable paradigm for constructing compact reasoning models by leveraging reasoning trajectories generated by powerful teachers.

Most existing reasoning distillation methods follow an offline training paradigm, where student models are optimized to imitate fixed reasoning trajectories generated by a teacher model. Despite their empirical success, these approaches suffer from a fundamental limitation. Teacher-generated reasoning traces often contain long and complex reasoning chains that exceed the learning capacity of the target student model. As a result, students are frequently forced to imitate reasoning patterns that they cannot effectively internalize, leading to inefficient knowledge transfer and limited reasoning generalization.

To address this issue, recent studies have explored online distillation frameworks that allow student models to generate trajectories under their own policies and improve through teacher feedback or preference-based optimization. Compared with offline imitation learning, online exploration can partially alleviate the distribution mismatch introduced by teacher forcing and encourage the development of autonomous reasoning behaviors. However, online distillation introduces a different challenge. Once the student deviates from a promising reasoning trajectory, errors may accumulate throughout subsequent reasoning steps, resulting in unstable exploration and degraded learning efficiency. Existing approaches typically rely on global reward signals or final-answer supervision, making it difficult to identify and correct mistakes at intermediate reasoning states.

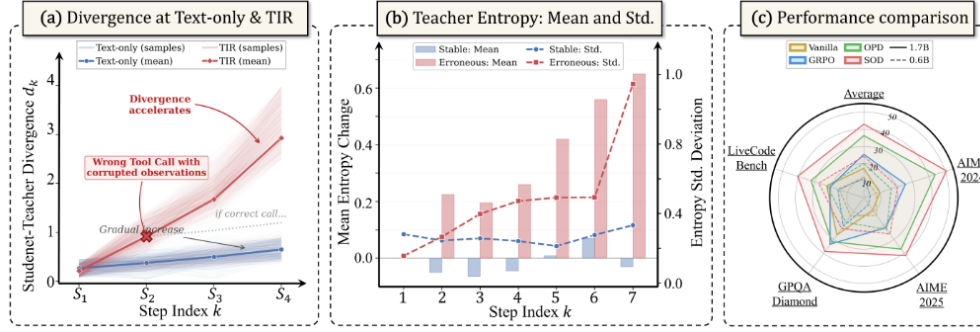


Figure 1: After completing the experiments, we should design a similar diagram showcasing the motivation, findings, and main results.

In this work, we view these limitations through a unified perspective. Effective reasoning transfer requires adapting both the training data and the learning process to the capability boundary of the student model. On the one hand, reasoning trajectories should be organized according to the student’s ability to absorb and understand them. On the other hand, the student should be able to recover from local reasoning failures during exploration rather than being constrained to fixed teacher demonstrations or delayed global feedback.

Motivated by this observation, we propose a capability-aware reasoning distillation framework consisting of an offline data adaptation stage and an online self-correction stage. In the offline stage, we introduce a student-perplexity-aware data curation pipeline that estimates the relative difficulty of teacher-generated reasoning trajectories from the perspective of the target student model. Based on these difficulty estimates, the curated corpus is organized into a structured curriculum while low-information reasoning segments are filtered out. In the online stage, we propose a teacher-anchored local alignment mechanism that monitors reasoning progress at the step level and performs localized rollback when the current trajectory enters potentially erroneous states. This enables the student model to autonomously re-explore alternative reasoning paths within its own capability range without relying on reinforcement learning.

Through the combination of student-aware data construction and teacher-anchored self-correction, our framework establishes a more effective reasoning transfer process for compact reasoning models. Extensive experiments on mathematical reasoning, medical question answering, and tool-integrated reasoning benchmarks demonstrate consistent improvements over conventional offline and online distillation baselines.

In summary, our main contributions are threefold:

- **Data-Centric CoT Curation:** We design a structured Chain-of-Thought (CoT) data pipeline that is aware of student perplexity. The pipeline leverages the conditional perplexity of the target student model to inversely quantify the difficulty of the corpus, thereby filtering out redundancies with low information content and organizing the dataset into a curriculum of three tiers: easy, medium, and hard. By operating exclusively on the data side, this design establishes an offline data foundation of high quality and cohesion, which is crucial for stabilizing the subsequent on-policy distillation on the method side.
- **Teacher-Anchored Autonomous Alignment:** We propose a novel local online alignment framework that empowers student models with endogenous self-correction capabilities, overcoming the rote-memorization flaw of standard distillation. By utilizing anchor-based state tracking, the system detects local cognitive dissonance and triggers lossless state roll-backs, forcing the student to autonomously explore multi-path solutions. Furthermore, we formulate a Hierarchical Preference Alignment Loss that decouples the optimization objectives: a Self-DPO term reinforces the student’s autonomously discovered correct logic, while a teacher-reference soft regularization term gently elevates the reasoning upper bound. This completely on-policy, RL-free approach achieves significant breakthroughs in both reasoning capacity and error tolerance.

- **Extensive Evaluation and Open-Source Release:** We conduct comprehensive evaluations across multiple high-complexity reasoning domains, including mathematical problem-solving and professional knowledge benchmarks. Empirical results demonstrate that lightweight student models trained via our framework significantly outperform baseline models distilled through standard KD and offline preference optimization, exhibiting superior generalization and reasoning stability. To facilitate reproducible research and foster community development in efficient LLM alignment, we open-source the complete structured data curation pipeline, the Teacher-Anchored online training framework, and the series of distilled model checkpoints.

2 PRELIMINARY

Offline Knowledge Distillation. In the alignment and model compression of Large Language Models (LLMs), Knowledge Distillation (KD) serves as a core paradigm. Given a highly capable pre-trained teacher model T and a student model π_θ to be optimized, traditional offline distillation relies on a static dataset generated by the teacher, denoted as $\mathcal{D}_T = \{(x, y) \mid x \sim \mathcal{X}, y \sim T(\cdot|x)\}$. The optimization objective is to minimize the negative log likelihood (NLL) of the student model over these fixed trajectories:

$$\min_{\theta} \mathcal{L}_{KD}(\theta) = \mathbb{E}_{(x, y) \sim \mathcal{D}_T} \left[- \sum_{t=1}^{|y|} \log \pi_{\theta}(y_t \mid x, y_{<t}) \right] \quad (1)$$

Mathematically, this is equivalent to minimizing the forward KL divergence under the empirical data distribution. However, this paradigm suffers from a severe distribution shift. In actual inference, the autoregressive generation distribution of the student $\hat{y} \sim \pi_{\theta}(\cdot|x)$ often diverges from the teacher-forcing distribution used during training. When complex chain-of-thought (CoT) reasoning is involved, forcing the student to fit trajectories that exceed its parameter capacity not only induces exposure bias but also fails to elicit its intrinsic reasoning potential.

On-Policy Exploration as MDP. To bridge the distribution gap between the training and inference stages, On-Policy Distillation (OPD) frames the text generation process as a reinforcement learning (RL) problem based on a Markov Decision Process (MDP). Given an input prompt $x \sim \mathcal{X}$, the state is defined as the current context $s_t = (x, y_{<t}) \in \mathcal{S}$, and the action space is the vocabulary $a_t = y_t \in \mathcal{V}$. The student model acts as an agent, actively exploring under its current policy $\pi_{\theta}(a_t|s_t)$ to maximize the expected reward $R_{teacher}$ provided by the teacher model.

To prevent policy degradation during exploration, a reverse KL divergence penalty based on a reference model π_{ref} is typically introduced:

$$\max_{\theta} J(\theta) = \mathbb{E}_{x \sim \mathcal{X}, y \sim \pi_{\theta}(\cdot|x)} [R_{teacher}(x, y) - \beta \mathbb{D}_{KL}(\pi_{\theta}(\cdot|x) \parallel \pi_{ref}(\cdot|x))] \quad (2)$$

where β is the temperature coefficient controlling the exploration boundary. Traditionally, this objective is optimized using policy gradient algorithms (e.g., PPO), which drive parameter updates via the token-level generalized advantage estimator $\hat{A}_t$.

However, in long-horizon complex reasoning, the estimation of the generalized advantage function ($\hat{A}_t$) is highly unstable. Once the lightweight student model deviates from the correct logical path, the subsequently generated context state s_{t+k} irreversibly drifts into an out-of-distribution (OOD) space. At this point, policy gradients based on delayed global rewards introduce massive variance and toxic gradient noise, leading to severe cascading errors and catastrophic forgetting of structural reasoning capabilities.

Direct Preference Optimization. To circumvent the high variance and instability introduced by explicit reward modeling and policy gradients, Direct Preference Optimization (DPO) proposes an RL-free alignment paradigm. Under the Bradley-Terry (BT) model assumption, the preference probability that trajectory y^+ is superior to y^- can be expressed by the implicit reward difference: $p(y^+ \succ y^-) = \sigma(R(x, y^+) - R(x, y^-))$.

By analytically solving the KL-constrained RL objective, DPO equivalently maps the reward function to the optimal policy as $R(x, y) = \beta \log \frac{\pi_\theta(y|x)}{\pi_{ref}(y|x)} + Z(x)$, thereby transforming the high-variance expected reward maximization problem into a more stable supervised classification loss:

$$\mathcal{L}_{DPO}(\theta) = -\mathbb{E}_{(x, y^+, y^-)} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y^+|x)}{\pi_{ref}(y^+|x)} - \beta \log \frac{\pi_\theta(y^-|x)}{\pi_{ref}(y^-|x)} \right) \right] \quad (3)$$

This RL-free paradigm significantly enhances training stability. Nevertheless, standard DPO strictly relies on global trajectory-level preference pairs (y^+, y^-) , which is mathematically equivalent to uniform credit assignment across the entire long reasoning chain. For high-complexity mathematical or agentic tasks, this coarse-grained global alignment fails to provide fine-grained correction signals for local deviations at intermediate logical states s_t . This fundamental mathematical and mechanistic flaw constitutes the primary motivation for our proposed endogenous lossless rollback and hierarchical preference alignment at the discrete local block level.

3 METHODOLOGY

3.1 STUDENT-PERPLEXITY-AWARE STRUCTURED DATA CURATION

Adopting a student-centric perspective, we leverage the untrained student base model, denoted as π_θ , to inversely quantify the knowledge discrepancy between the student and the teacher models. For a given reasoning trajectory $y = (y_1, y_2, \dots, y_T)$ generated by the teacher model conditioned on prompt x , we compute the token-level conditional perplexity (PPL) of the student model given its preceding context:

$$\text{PPL}(y_t) = \exp(-\log \pi_\theta(y_t | x, y_{<t})) \quad (4)$$

To evaluate the relative cognitive difficulty of the entire sample with respect to the target student model, we define the trajectory-level difficulty score, $S(x, y)$, as the average log-perplexity over the set of core logical tokens, denoted as $\mathcal{C}$:

$$S(x, y) = \frac{1}{|\mathcal{C}|} \sum_{t \in \mathcal{C}} \log \text{PPL}(y_t) = -\frac{1}{|\mathcal{C}|} \sum_{t \in \mathcal{C}} \log \pi_\theta(y_t | x, y_{<t}) \quad (5)$$

Concurrently, during the computation of token-level PPL, our pipeline automatically identifies and filters out low-information redundancies. Tokens for which the student model assigns an extremely high prior probability and an exceptionally low PPL (e.g., generic transitional phrases such as “Therefore” or “As a result”) are classified as stylistic redundancies that do not contribute to the student’s logical reasoning. The pipeline dynamically compresses or penalizes this redundant information at the data level, thereby preventing the student model from expending its limited parameter capacity on merely mimicking the teacher’s expressive style during subsequent training.

Based on the distribution of the trajectory difficulty scores $S(x, y)$, the pipeline automatically partitions the refined corpus into three mutually disjoint difficulty strata using dual thresholds (τ_1 and τ_2) to establish a progressive curriculum structure. Specifically, the **Foundation Consolidation Set** ($\mathcal{D}_{\text{easy}}$) satisfies $S(x, y) < \tau_1$ and contains fundamental conventions that are readily comprehensible to the student. The **Zone of Proximal Development Set** ($\mathcal{D}_{\text{mid}}$) satisfies $\tau_1 \leq S(x, y) \leq \tau_2$ and serves as the golden samples for core logical transfer across domains. Finally, the **Hardcore Challenge Set** ($\mathcal{D}_{\text{hard}}$) satisfies $S(x, y) > \tau_2$ and represents high-difficulty, long-tail reasoning tasks.

This curriculum arrangement, executed exclusively at the data level, provides a robust backbone for smooth, low-variance convergence during the subsequent offline pre-alignment phase of the student model.

3.2 TEACHER-ANCHORED STATE TRACKING AND LOSSLESS ROLLBACK

The primary bottleneck of traditional On-Policy Distillation (OPD) paradigms in long-horizon, complex reasoning tasks lies in the vulnerability to error propagation during autoregressive decoding. Formally, given an input prompt $x \in \mathcal{X}$, standard OPD operates over the entire sampled trajectory $y \sim \pi_\theta(\cdot | x)$ by supplying token-level advantage signals as supervision. However, once the

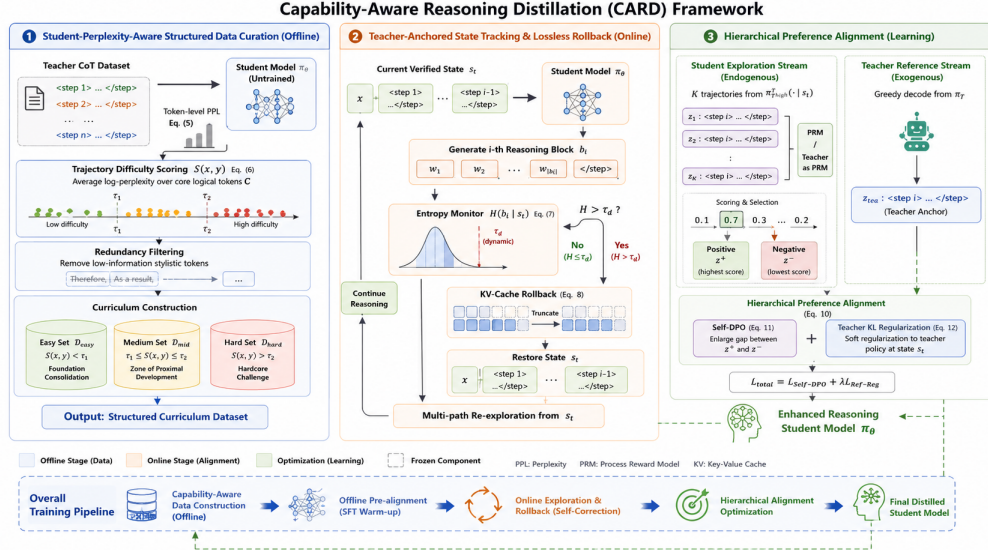


Figure 2: Main experimental results and comprehensive analysis integrated across the paper, highlighting our key findings and the core challenges addressed in this work.

lightweight student policy π_θ deviates from the gold-standard reasoning path at any intermediate time-step t , the subsequent context state $c_{T>t} \triangleq (x, y_{<t}, y_{t:T})$ irreversibly drifts into the out-of-distribution (OOD) space. Forcing a powerful large language model (LLM) teacher $\pi_T(\cdot | c_T)$ to provide conditional probability guidance over such heavily corrupted semantic contexts induces a severe surge in conditional entropy. The resulting chaotic supervision signals act as toxic gradients that contaminate the student’s network weights, which subsequently leads to severe training instability and the catastrophic forgetting of structural reasoning capabilities.

To circumvent this supervision failure induced by state drift, we propose a novel teacher-anchored state tracking and lossless rollback mechanism from the endogenous cognitive perspective of the student model. We leverage structured semantic anchors, specifically `<step>` and `</step>` tokens injected during the offline data preparation phase, to explicitly demarcate the boundaries of local reasoning states within the Markov decision process. Let the currently verified correct context state be denoted as $s_t = (x, b_1, b_2, \dots, b_{i-1}) \in \mathcal{S}$, where $b_{<i}$ represents the sequence of the first $i-1$ validated logical blocks. When the student policy π_θ autoregressively generates the i -th local reasoning block $b_i = (w_1, w_2, \dots, w_{|b_i|})$ from the state s_t and outputs the closing anchor `</step>`, the system evaluates its endogenous local uncertainty by computing the average token-level information entropy $H(b_i | s_t)$ over the generated block interval:

$$H(b_i | s_t) = -\frac{1}{|b_i|} \sum_{j=1}^{|b_i|} \sum_{w \in \mathcal{V}} \pi_\theta(w | s_t, w_{<j}) \log \pi_\theta(w | s_t, w_{<j}) \quad (6)$$

where $\mathcal{V}$ denotes the discrete token vocabulary. We introduce a difficulty-adaptive dynamic threshold $\tau_d = \Phi(\mathcal{M}(\tau))$, which is monotonically mapped from the global trajectory difficulty score $\mathcal{M}(\tau)$ computed in the data curation phase.

Crucially, a high localized entropy indicates the student’s endogenous uncertainty—often caused by multiple viable reasoning paths—rather than an absolute logical failure. To prevent the erroneous penalization of valid exploration, we introduce a *Confidence-Triggered Lazy Verification* mechanism. If the localized entropy satisfies $H(b_i | s_t) > \tau_d$, the system detects a potential hallucination and raises a *cognitive suspicion flag*, temporarily suspending the student’s autoregressive generation. At this critical juncture, the teacher model π_T intervenes on-demand to perform a rigorous logical evaluation of the suspect block b_i . Let $\mathcal{V}_T(b_i | s_t) \in \{0, 1\}$ denote the binary verification outcome provided by the teacher, where 0 indicates a logical deviation. If the block is logically sound ($\mathcal{V}_T = 1$), it is accepted as a valid exploratory path, and the generation resumes.

However, if the teacher confirms a logical collapse ($\mathcal{V}_T = 0$), the system triggers an immediate cognitive dissonance signal. In contrast to conventional reinforcement learning frameworks that passively penalize suboptimal paths via delayed global rewards—which typically suffer from astronomical variance and sample inefficiency—our framework intercepts the error exactly at the point of confirmation. Instead of continuing training on the corrupted context or blindly applying global teacher-forcing, the underlying inference framework executes a localized, lossless rollback. By interfacing directly with the memory management layers of the distributed inference engine, such as PagedAttention, the system physically truncates the Key-Value Cache (KV-Cache) of the sequence, forcing the computation graph to revert precisely to the temporal boundary of state s_t :

$$\text{KV-Cache}(\pi_\theta) \leftarrow \text{Truncate}(\text{KV-Cache}(\pi_\theta), s_t) \quad (7)$$

By entirely purging the flawed block b_i from both the physical memory and the historical context, we effectively obstruct the cascading amplification of early errors in both spatial and temporal dimensions. This localized stop-loss paradigm guarantees a pristine and deterministic context for the student to re-invoke its parameter space for local multi-path exploration and online correction.

3.3 HIERARCHICAL PREFERENCE ALIGNMENT VIA AUTONOMOUS EXPLORATION

Upon successfully resetting the context to the deterministic state s_t via the lossless rollback mechanism, the framework concurrently initiates a dual-stream parallel sampling strategy at this critical bifurcation junction, capturing both the maximum endogenous potential of the student and the absolute spatial upper bound of the teacher. In the first endogenous stream, we introduce explicit temperature perturbations to amplify the student policy’s exploratory capacity within the local solution space. Formally, we draw K independent trajectories from the perturbed student distribution $\pi_{\theta, T_{\text{high}}}(\cdot | s_t)$ to construct a localized candidate set $Z_{\text{stu}} = \{z_1, z_2, \dots, z_K\}$. We then deploy a process reward model (PRM) or utilize the teacher model as a localized verification mapping function $\mathcal{R} : \mathcal{S} \times \mathcal{Z} \rightarrow \mathbb{R}$ to formalize and score the logic of each candidate block. The logically closed and highest-scoring path is designated as the positive anchor z^+ , while the lower-scoring or logically broken paths are categorized as negative anchors z^- :

$$z^+ = \arg \max_{z_k \in Z_{\text{stu}}, \mathcal{R}(s_t, z_k) > 0} \mathcal{R}(s_t, z_k), \quad z^- = \arg \min_{z_k \in Z_{\text{stu}}} \mathcal{R}(s_t, z_k) \quad (8)$$

Crucially, because the positive anchor z^+ originates entirely from the student’s own conditional probability distribution $\pi_\theta(\cdot | s_t)$, it represents the optimal solution that the student can naturally comprehend and replicate within its current cognitive capacity. Simultaneously, the second exogenous stream initiates a single greedy decoding step from the identical state s_t using the teacher model π_T to generate a standard reference block $z_{\text{tea}} \sim \arg \max_z \pi_T(z | s_t)$ ex post facto. In this collaborative formulation, the teacher avoids invasive intervention in the student’s autoregressive generation process, serving instead as an external frame of reference that provides a localized supremum within the high-dimensional semantic space.

To fundamentally eliminate the training instabilities and gradient divergence caused by the capacity mismatch between the oversized teacher and the lightweight student, we abandon standard token-level forward cross-entropy loss. Instead, we formulate a Hierarchical Preference Alignment Loss that decouples the optimization objectives into endogenous self-correction and exogenous soft regularization. The total optimization objective is expressed as a linear combination of the self-correction preference loss $\mathcal{L}_{\text{Self-DPO}}(\theta)$ and the teacher-reference regularization term $\mathcal{L}_{\text{Ref-Reg}}(\theta)$:

$$\mathcal{L}_{\text{Total}}(\theta) = \mathcal{L}_{\text{Self-DPO}}(\theta) + \lambda \mathcal{L}_{\text{Ref-Reg}}(\theta) \quad (9)$$

where $\lambda \in \mathbb{R}^+$ is a hyperparameter balancing the internal and external optimization driving forces. The first component applies the Direct Preference Optimization (DPO) mathematical framework at the discrete block level, enlarging the relative log-likelihood gap between the student’s self-discovered positive anchor z^+ and negative anchor z^- :

$$\mathcal{L}_{\text{Self-DPO}}(\theta) = -\mathbb{E}_{s_t \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(z^+ | s_t)}{\pi_{\text{ref}}(z^+ | s_t)} - \beta \log \frac{\pi_\theta(z^- | s_t)}{\pi_{\text{ref}}(z^- | s_t)} \right) \right] \quad (10)$$

where π_{ref} denotes the frozen baseline student policy initialized via supervised fine-tuning (SFT) during the cold-start phase, $\sigma(\cdot)$ is the standard sigmoid function, and β serves as the KL-divergence coefficient penalizing policy drift. Backpropagating the gradients of $\mathcal{L}_{\text{Self-DPO}}(\theta)$ forces the student

to deeply internalize its localized successes and failures, thereby robustly enhancing its endogenous error-recovery capabilities.

Although the student’s autonomously discovered positive anchor z^+ is logically viable, its expressive compactness, mathematical rigor, and high-dimensional representations typically lag behind those of the teacher. To prevent the student from converging toward low-level logical equilibrium and to continuously elevate its reasoning upper bound, we incorporate the second exogenous soft regularization term $\mathcal{L}_{\text{Ref-Reg}}(\theta)$. This term computes the forward Kullback-Leibler (KL) divergence between the teacher and student policies at the token sequence level under state s_t , imposing a smooth geometric constraint on the student’s behavior space:

$$\mathcal{L}_{\text{Ref-Reg}}(\theta) = \mathbb{E}_{s_t \sim \mathcal{D}} [D_{\text{KL}}(\pi_T(\cdot | s_t) \parallel \pi_\theta(\cdot | s_t))] = \mathbb{E}_{s_t \sim \mathcal{D}} \left[\sum_{w \in \mathcal{V}} \pi_T(w | s_t) \log \frac{\pi_T(w | s_t)}{\pi_\theta(w | s_t)} \right] \quad (11)$$

This entirely on-policy, RL-free design eliminates the need for complex multi-stage alignment pipelines, ensuring that the student’s optimization trajectory remains firmly anchored within its endogenous vocabulary distribution while being smoothly regularized by the teacher’s high-dimensional probability space.

4 EXPERIMENTS

4.1 SETUP

Training Data. To comprehensively validate the effectiveness of our approach across complex reasoning modalities, we construct structured reasoning corpora spanning three core domains: mathematics, medicine, and agent-based tool-use. To seamlessly integrate with the student-perplexity-aware structured curation and state rollback mechanisms proposed in Section 3, we enforce uniform formatting constraints during the data generation phase. Specifically, we deploy a strong teacher model (Qwen3-32B) to generate Chain-of-Thought (CoT) trajectories for all samples, prompting the teacher model to dynamically inject `<step>` and `</step>` semantic anchors during generation as explicit boundaries for logical blocks. These structured anchors not only provide the basis for calculating local cognitive difficulty (i.e., the trajectory-level perplexity) in the offline phase but also establish precise truncation and rollback coordinates for the online alignment phase detailed in Section 3.2. The specific data construction details for each domain are as follows:

Mathematics Domain: We construct our training set based on the large-scale mathematical reasoning dataset, OpenMathReasoning. We deploy Qwen3-32B as the teacher model to re-solve the problems and reconstruct the CoT trajectories. During generation, the teacher model is instructed to strictly encapsulate each step of the mathematical derivation within `<step>` tags. Given the massive scale of the original dataset, we filter out samples exhibiting generation failures or formatting errors to yield a high-quality subset for our experiments.

Medical Domain: To prevent data contamination, we rigorously exclude all test cases present in our evaluation benchmarks. We utilize the official training splits of MedQA and MedMCQA as our raw corpora. Given the critical reliance on strict logical chains in medical diagnostics, we prompt the teacher model to employ `<step>` tags to clearly demarcate distinct reasoning stages: patient symptom extraction, pathological analysis, differential diagnosis, and the final conclusion. This step-wise structuring is vital for medical tasks, where a single logical flaw in intermediate pathology reasoning can easily lead to a severe diagnostic deviation, making our localized rollback mechanism particularly impactful.

Agent Domain: For Tool-Integrated Reasoning (TIR) tasks, we utilize the standard training sets from ToolBench, supplemented by APIBench. In interactive agent environments, cascading errors are particularly detrimental—a single hallucinated API call or incorrect parameter can cause a lightweight student model to enter an irrecoverable failure loop. Consequently, when the teacher model generates interaction trajectories, the `<step>` tags are precisely positioned to encapsulate each complete “Thought $\rightarrow$ Action $\rightarrow$ Action Input” closed loop. This chunking strategy effectively isolates anomalous states triggered by external environment returns, providing a robust structural foundation for filtering out spurious gradients during the subsequent online training.

Cold start Before the online preference alignment phase, we apply a brief supervised fine tuning to initialize the base student model π_θ and establish a stable reference policy π_{ref} . This step equips the student with basic structured reasoning skills while ensuring gradient stability during early online exploration. Specifically, we train the model using the standard negative log likelihood loss on the curated offline data described in Section 3.1. To preserve the internal exploration capacity of the student and prevent overfitting to the complex generation patterns of the teacher, we strictly limit this stage to a single epoch. This lightweight initialization enables the student to reliably output reasoning trajectories with $\langle \text{step} \rangle$ and $\langle / \text{step} \rangle$ boundary tags, which serves as the essential formatting prerequisite for triggering the subsequent localized state rollback mechanism.

Evaluation Benchmark To comprehensively evaluate the generalizability and robustness of our approach across different modalities, we conduct extensive experiments on a diverse set of benchmarks spanning three complex reasoning domains: **Mathematical Reasoning**. We evaluate on AIME25, AIME24, AMC2023, OlympiadBench, and GSM8K, covering a wide range of difficulty levels and reasoning styles. Following standard practice, we report the pass rate with 16 samples (Avg@16) for AIME25, AIME24, and AMC2023 due to the limited number of test problems. For OlympiadBench and GSM8K, we report the zero-shot Pass@1 accuracy; **Medical Domain**. To assess domain-specific logical alignment and professional knowledge reasoning, we evaluate our models on HealthBench-500, LLMEval-Med, MedQA, and MedMCQA. For these benchmarks, we report the standard zero-shot accuracy; **Tool-Integrated Reasoning**. To test the model’s tolerance against cascading errors in interactive environments, we utilize BFCL, ToolBench, APIBench, and T-Eval. We report the successful execution rate and valid tool-calling accuracy as defined by their respective official evaluation protocols.

Models Across the three evaluation domains (mathematical, medical, and agent reasoning), we employ a consistent model setup. We select Qwen3-32B as the teacher model because of its robust reasoning capabilities and its proficiency in generating detailed chain of thought trajectories. To rigorously evaluate the effectiveness of our distillation framework on highly compact architectures, we utilize Qwen3-0.6B as the target student model.

Baseline Methods In math task, we compare against four baseline methods: (1) **Standard KD**. Direct distillation from the teacher’s long CoT sequences via rejection sampling. To ensure fairness, we evaluate both the base student and the cold-started student under this setting. (2) **Self-Distillation**. The student performs rejection sampling on their own generated data. In this case, we use the cold-started student as the initial model. (3) **MoRSD**, which distills CoT using a self-guided trajectory difficulty metric to select high-quality trajectories for efficient student training. (4) **Gen-SSD**. A generation-time self-selection method where the student evaluates and prunes candidate continuations during the teacher’s reasoning generation.

4.2 MAIN RESULTS

4.3 ABLATION STUDIES

To comprehensively verify the empirical inner-workings and structural necessity of each constituent module within our Capability-Aware Reasoning Distillation (CARD) framework, we execute an extensive battery of ablation experiments. We evaluate several distinct architectural and algorithmic variants of our Qwen3-0.6B student model by systematically deactivating or mutating core mechanisms across both the offline data engineering phase and the online reinforcement-learning-free alignment phase.

Component-Level Performance Breakdown. We first evaluate the macro-level synergy of the four primary pillars of CARD: Student-Perplexity-Aware Curation, Teacher-Anchored Local KV-Cache Rollback, Block-Level Self-DPO, and Teacher-Reference Regularization ($\mathcal{L}_{Ref-Reg}$). Table 2 provides a granular multi-domain performance breakdown. Removing the offline PPL curation pipeline results in immediate generalization drop across all datasets, confirming that raw, uncured teacher traces induce exposure bias. Crucially, disabling the Local Rollback mechanism triggers a catastrophic performance drop in interactive domains such as ToolBench, validating our claim that global, trajectory-level rewards are insufficient for mitigating intermediate cascading errors. The op-

Table 1: Comprehensive performance comparison across Mathematical, Medical, and Tool-Integrated Reasoning benchmarks. We report zero-shot accuracy or specific benchmark metrics. Best results among student models are highlighted in bold.

Benchmark	Teacher	0.6B Base	Logit-KD	Self-Distil	MoRSD	Gen-SSD	Ours
<i>Domain: Mathematical Reasoning</i>							
AIME26	70.42	11.25	9.17	8.54	-	-	-
AIME25	72.90	18.54	11.46	15.83	-	-	-
Omni-MATH	58.80	24.80	21.60	27.40	-	-	-
OlympiadBench	72.85	38.58	29.53	37.69	-	-	-
Math Avg.	-	-	-	-	-	-	-
<i>Domain: Medical Reasoning</i>							
HealthBench-500	-	-	-	-	-	-	-
LLMEval-Med	-	-	-	-	-	-	-
MedQA	-	-	-	-	-	-	-
MedMCQA	-	-	-	-	-	-	-
Med Avg.	-	-	-	-	-	-	-
<i>Domain: Tool-Integrated Reasoning (Agent)</i>							
BFCL	-	-	-	-	-	-	-
ToolBench	-	-	-	-	-	-	-
APIBench	-	-	-	-	-	-	-
T-Eval	-	-	-	-	-	-	-
Agent Avg.	-	-	-	-	-	-	-

timel coordination of all modules yields the highest absolute gains, proving they are complementary rather than redundant.

Table 2: Granular ablation study breaking down the exact performance contributions of CARD’s individual core modules across distinct benchmarks in the Mathematical, Medical, and Tool-Integrated Reasoning domains. All training variants use the identical Qwen3-0.6B student backbone, and all numeric entries are initialized as placeholders (— · —). The bold entries denote the optimal paradigm matching our complete framework.

Offline Cur.	KV-Rollback	Self-DPO	Ref-Reg	Mathematical Reasoning		Medical Reasoning		Tool-Integrated Agent	
				AIME 24	Omni-MATH	MedQA	MedMCQA	BFCL	ToolBench
—	—	—	—	— · —	— · —	— · —	— · —	— · —	— · —
✓	—	—	—	— · —	— · —	— · —	— · —	— · —	— · —
✓	✓	—	—	— · —	— · —	— · —	— · —	— · —	— · —
✓	—	✓	✓	— · —	— · —	— · —	— · —	— · —	— · —
✓	✓	✓	—	— · —	— · —	— · —	— · —	— · —	— · —
✓	✓	✓	✓	— · —	— · —	— · —	— · —	— · —	— · —

Impact of Curriculum Curation and Dynamic Data Blending. As formulated in Section 3.1, our offline data foundation is stratified into three mutually disjoint subsets ($\mathcal{D}_{easy}$, $\mathcal{D}_{mid}$, $\mathcal{D}_{hard}$) via dynamic thresholds. We investigate the impact of different blending ratios during the cold-start supervised fine-tuning (SFT) stage. Table 3 highlights the performance shifts under various curriculum ratios. Shifting the data composition drastically toward the hardcore challenge set (5% + 15% + 80%) triggers early gradient destabilization and poor convergence, as the traces far exceed the initial parameter capacity of the 0.6B student. Conversely, an over-reliance on basic conventions (70% + 20% + 10%) yields a student model that fails to unlock advanced reasoning paths. The

empirically derived balanced distribution (30% + 50% + 20%) serves as the optimal mathematical baseline.

Table 3: Ablation analysis evaluating the effects of different dynamic data blending ratios among the Foundation Consolidation ($\mathcal{D}_{easy}$), Zone of Proximal Development ($\mathcal{D}_{mid}$), and Hardcore Challenge ($\mathcal{D}_{hard}$) datasets during the cold-start pre-alignment phase. Performance metrics represent the averaged domain accuracy across Math, Medicine, and Agent tasks.

Easy Ratio (%)	Medium Ratio (%)	Hard Ratio (%)	Math Avg.	Med Avg.	Agent Avg.
70	20	10	— . —	— . —	— . —
20	60	20	— . —	— . —	— . —
5	15	80	— . —	— . —	— . —
30	50	20	— . —	— . —	— . —

Step-Level Error Mitigation and State Divergence Dynamics. To isolate the physical impact of the Teacher-Anchored Local Alignment mechanism, we inspect the micro-level context dynamics during the online exploration phase. Figure 3 tracking the token-level state divergence (d_k) across multiple reasoning step indices k . Without the Confidence-Triggered Lazy Verification and subsequent KV-cache truncation, early logical deviations amplify exponentially, leading the context to slide irreversibly into out-of-distribution (OOD) territory. By physically purging the flawed memory blocks from PagedAttention layers, CARD suppresses the conditional entropy surge of the teacher supervision, ensuring a highly stable and bounded gradient trajectory throughout the entire long-horizon inference chain.



Figure 3: Dynamic tracking of step-level student-teacher semantic state divergence (d_k) over long-horizon trajectories. The subplots contrast the baseline online exploration model (without rollback, showing exponential error growth) against the CARD framework (Ours, showing tightly bounded divergence due to physical state truncation) across the Mathematical, Medical, and Agent domains respectively.

Granularity of Local Verification Blocks. The boundary demarcation of local reasoning states represents a core design choice. We explore the optimal chunking granularity of the reasoning blocks (b_i) by altering the anchor injection intervals. We experiment with three configurations: (1) *Token-level verification*, where entropy is calculated per token; (2) *Sentence-level verification*, utilizing standard punctuation; and (3) *Anchor block-level verification* (Ours), which follows the teacher’s logical sub-steps ($\langle step \rangle$). As visualized in Figure 4, token-level checking introduces extreme computational overhead and severe false-alarm rates, overly restricting valid student exploration. On the other hand, sentence-level grouping fails to capture atomic logical jumps in code or math derivations. Demarcating states via endogenous semantic steps strikes the optimal trade-off between exploration utility and verification precision.

Hyperparameter Sensitivity of Dual-Stream Alignment. Finally, we study the hyperparameter landscapes governing the Hierarchical Preference Alignment Loss in Equation (9) and (10). Fig-

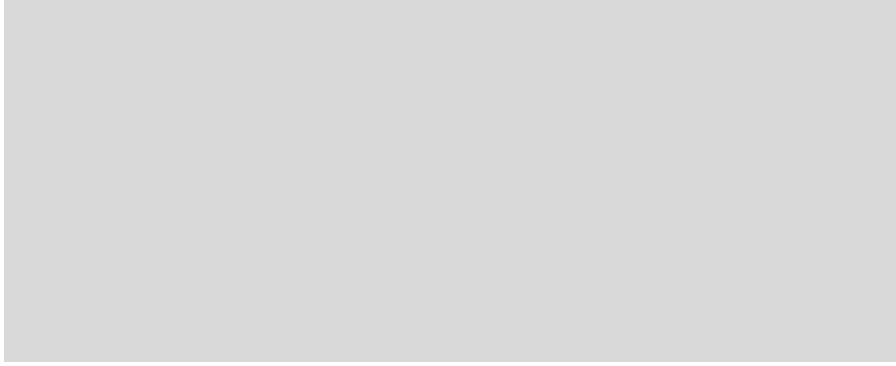


Figure 4: Comparative analysis of different verification and rollback granularities (Token-level, Sentence-level, and our semantic Anchor-step level). The left Y-axis indicates the benchmark pass rate, while the right Y-axis reflects the additional computational latency overhead induced by teacher interaction. Our anchor-step design achieves optimal performance with minimal inference delay.

Figure 5 illustrates the combined impact of the DPO consensus scaling coefficient β and the soft regularization balancing coefficient λ . We observe that setting β too small causes policy drift due to insufficient penalty on distribution shift, while excessively high values inhibit the student’s learning rate. Similarly, the optimal value of λ acts as a smooth geometric guide; a localized peak performance region emerges, demonstrating that a structured blend of internal self-correction and external teacher reference is paramount for scaling compact reasoning architectures.



Figure 5: Hyperparameter sensitivity landscape showing the joint effect of the KL-divergence penalty β and the teacher soft regularization coefficient λ on the final combined reasoning score. The 3D surface plot or 2D contour heatmap showcases that the performance peak is localized within a robust, centralized hyperparameter sweet spot.

5 RELATED WORK

CoT Distillation Data Construction The quality of reasoning data is a key factor for effective chain of thought distillation. While early approaches rely on supervised fine tuning over large collections of teacher generated trajectories, these traces often contain highly redundant steps and overly complex reasoning patterns, leading to inefficient knowledge transfer. To address this, recent data centric studies have explored filtering by correctness, reward scores, and trajectory quality, or organizing training samples via curriculum learning and student aware difficulty estimation. In contrast to prior work, we characterize the knowledge gap between the teacher trajectory and the target student using conditional perplexity measured by the student model itself. Based on this endogenous signal, we construct a structured reasoning curriculum and dynamically filter out low information segments, establishing a highly coherent data foundation for subsequent online distillation.

Online Distillation and Alignment Conventional offline knowledge distillation forces student models to imitate fixed teacher generated trajectories, often suffering from distribution mismatch and error accumulation during long horizon reasoning since training contexts diverge from autonomous inference. To alleviate these limitations, recent works have investigated online distillation and self improvement mechanisms, allowing students to generate trajectories under their current policies and optimize via teacher feedback. While related directions like reinforcement learning based distillation, preference optimization, and reflective self correction encourage autonomous exploration, they typically rely on global reward signals or delayed final outcome supervision. Unlike existing methods, we focus on correcting errors at intermediate reasoning states. We introduce a teacher anchored local alignment framework that tracks reasoning progress at the step level and enables localized rollback when the current trajectory deviates from a promising path. This mechanism empowers the student model to perform autonomous correction and exploration strictly within its own capability boundary, achieving highly stable online alignment without complex reinforcement learning.

6 CONCLUSION

In this paper, we proposed the Capability-Aware Reasoning Distillation (CARD) framework to address the critical challenges of distribution shift and error accumulation in complex reasoning distillation. By coupling student-perplexity-aware offline curriculum construction with a teacher-anchored online local alignment mechanism, our framework enables lightweight student models to effectively correct intermediate logical deviations through localized, lossless KV-cache rollbacks. Extensive experiments across mathematical, medical, and agent-based reasoning benchmarks demonstrate that CARD significantly outperforms traditional offline and online distillation baselines, achieving superior generalization and reasoning stability on a highly compact 0.6B student architecture.

Beyond validating the empirical effectiveness of our framework, our comprehensive evaluation also revealed several insightful findings regarding student capacity awareness and the structural optimization of teacher chain of thought trajectories. Future work will extend these findings to further explore the frontiers of on-policy distillation. Specifically, we aim to investigate adaptive mechanism designs that can dynamically modulate the reasoning granularity of the teacher model to precisely match the cognitive boundaries of scaling-limited student networks, thereby establishing a more robust and self-evolving alignment paradigm.

REFERENCES

A ADDITIONAL RELATED WORKS

Test time scaling Recently, the paradigm of test time scaling, which dynamically allocates additional computational resources during inference to unlock the complex logical reasoning potential of large language models, has emerged as a core research frontier. Current mainstream approaches primarily rely on global search algorithms guided by process reward models, such as beam search, Monte Carlo Tree Search, or best of N sampling based on majority voting. Although these exhaustive or heuristic sampling techniques for test time adaptation can significantly improve final reasoning accuracy, they often suffer from node explosion and exorbitant Key Value cache memory overhead in long horizon tasks. Furthermore, lightweight student models lacking external priors are highly susceptible to self reinforcing logical dead loops when exploring without constraints. To strike a balance between exploration utility and computational efficiency, our approach completely discards heavy global multipath expansion and introduces a lightweight, lossless local rollback mechanism. By continuously monitoring local information entropy during the autoregressive generation process at the foundational level, our system accurately truncates the physical memory of anomalous states the moment cognitive deviation is detected. It then initiates local bifurcation exploration strictly within the current deterministic boundary. Consequently, our method achieves more targeted and memory efficient state adaptation compared to traditional global tree search, without significantly increasing the overall inference computational budget.

Preference alignment In the alignment fine tuning of large language models, reinforcement learning from human feedback typically employs algorithms like PPO or the recently successful GRPO

for complex mathematical reasoning to maximize expected rewards. However, when faced with long sequence chain of thought tasks spanning thousands of tokens, standard reinforcement learning methods encounter severe reward sparsity, credit assignment difficulties, and massive training instability. Their online convergence on lightweight models relies heavily on hyperparameter tuning. To mitigate these high variance challenges, Direct Preference Optimization and its variants have pioneered an alignment paradigm free of reinforcement learning, greatly simplifying the training pipeline. Nevertheless, standard Direct Preference Optimization remains heavily dependent on trajectory level global preferences, making it difficult to provide fine grained guidance for intermediate steps in the reasoning chain. Although recent step level optimization attempts introduce step data, they are still limited by static offline annotations. Building upon these cutting edge works, we construct a hierarchical preference alignment loss based entirely on autonomous model exploration. At the local verification bifurcation point triggered after our lossless rollback, we dynamically sample to generate positive and negative cognitive anchors. We directly apply our self preference optimization at the discrete local block level to reinforce the correct logic discovered endogenously by the student model. Innovatively, we supplement this with soft regularization from the teacher model to smoothly elevate its exploration upper bound. This achieves adaptive and highly stable fine grained process alignment while entirely discarding complex reinforcement learning pipelines.

Online distillation Knowledge distillation is the core paradigm for achieving lightweight reasoning models, but traditional offline sequence level distillation forces students to fit fixed teacher trajectories, which easily triggers severe distribution shifts and exposure bias in complex reasoning tasks. To narrow the gap between training and inference stages, recent methods such as generalized knowledge distillation and online reflection have introduced the online distillation paradigm. This allows the student model to sample autonomously under its current policy and dynamically update based on real time rewards or token level Kullback Leibler divergence provided by the teacher model. However, online exploration faces a severe error cascade problem in long horizon reasoning. Once a lightweight student deviates from the correct logical path, its subsequently generated context irreversibly slides into a severe out of distribution space. At this point, forcibly utilizing the large teacher model to impose probability supervision on these contaminated catastrophic states is not only difficult for the student to digest but also introduces massive noise gradients, destroying the established reasoning structure of the student. Unlike these traditional online distillation methods that passively receive feedback, we propose a teacher anchored online local alignment framework that endows the student model with endogenous state awareness. Through active physical state truncation and history cleansing, we eradicate toxic gradients at their source, providing a permanently reliable and deterministic context prefix for the online exploration of the student. This constructs a self evolving distillation environment with extremely high robustness and fault tolerance.

B ALGORITHM PSEUDOCODE

C DETAILED EXPERIMENTAL SETTINGS

OpenMathReasoning Process raw data 192,523, after deduplication about 12,926, 57 without boxed. After deduplication, we further filter the data using HuggingFace’s math-verify to check the symbolic equivalence between each problem’s expected-answer and the final boxed answer in the generated solution. We discard the 1,800 samples flagged as non-equivalent. Although manual inspection shows that many of these are in fact mathematically equivalent (e.g., differing only in arbitrary constant names such as $f(x)=ax$ vs. $f(x)=kx$, dummy-variable renaming, attached natural-language qualifiers, units, or alternative algebraic forms), the surface form of the answer disagrees with the reference. Keeping such samples would expose the student model to inconsistent answer conventions during distillation and risk training it to produce ambiguous or non-canonical outputs. We therefore remove them, yielding a final clean set of **11,064** samples. Qwen3-32B correct rate:93.00(10,289/11,064), avg tokens:7,201, inference time:10h,44mins(3 node x 4 TP) only use 10,289 data

MathBenchmark AIME24:30 AIME25:30 AIME26:30 AMC23:40 OlympiadBench:674 GSM8K:1,319, turn to boxed

How to get golden path? DPO cold start? Use API to simple CoT?

Dynamic Blending 70+20+10,20+60+20,5+15+80(30,50,20)

D DETAILED EXPERIMENTAL ANALYSIS

Table 4: Comparison of Self-Correction Signal Density (per 1,000 tokens)

Signal Word	4B Density	32B Density	Ratio (4B/32B)
“Wait”	3.80	1.55	2.45×
“But wait”	1.20	0.41	2.93×
“Therefore”	9.50	3.51	2.71×
“Alternatively”	2.10	0.20	10.50×
“Let me check”	1.60	1.62	~1.0×
“Hmm”	0.30	0.00	–

E PROMPT TEMPLATE

Table 5: Prompt for Golden Path Chain-of-Thought Distillation

System Prompt

You are an expert mathematician and a senior reasoning-trace curator who builds high-quality chain-of-thought (CoT) distillation data for training mathematical reasoning models.

You will be given:

- **PROBLEM**: a mathematics problem.
- **VERIFIED_ANSWER**: the ground-truth final answer (already verified to be correct).
- **DRAFT_REASONING**: a raw, exploratory CoT produced by a teacher model. It reaches the correct answer, but it is NOT a golden path: it contains backtracking, false starts, self-doubt ("wait", "hmm", "let me reconsider", "actually"), redundant restatements, abandoned branches, and meandering exploration.

Your task is to transform DRAFT_REASONING into a GOLDEN PATH: the single, clean, optimal reasoning trajectory.

Strict requirements:

1. **LINEAR & MONOTONIC**: every step must move forward toward the answer. Remove all backtracking, dead ends, abandoned attempts, and self-correction loops. Keep only the steps that actually contribute to the correct solution.
2. **NO FILLER**: eliminate hedging and meta-talk such as "wait", "hmm", "let me think", "maybe", "actually", "I wonder". Replace exploratory phrasing with confident, deliberate reasoning.
3. **RIGOROUS & COMPLETE**: do not skip any logically necessary step. The chain must be fully justified and self-contained; a reader should be able to follow it without gaps.
4. **FAITHFUL**: stay true to the mathematics and, where sound, to the solution strategy already present in DRAFT_REASONING. Do NOT invent facts, theorems, or numbers. Do NOT alter the problem.
5. **CORRECT ENDPOINT**: the reasoning must conclude with the final answer, and that answer MUST equal VERIFIED_ANSWER.
6. **STYLE** — match the original teacher trace: write as a continuous, first-person internal monologue in flowing paragraphs ("I need to... So... Since... Therefore..."), exactly like the DRAFT_REASONING voice but cleaned of all detours. Do NOT add section headers, bullet outlines, horizontal rules, or a textbook-style layout. Keep it as natural running reasoning, just shorter and strictly forward-moving.
7. **FORMAT**: output ONLY the golden-path reasoning. End with the final answer in the form `\boxed{<final answer>}`, where the boxed value equals VERIFIED_ANSWER. Do not include the labels PROBLEM/VERIFIED_ANSWER/DRAFT_REASONING, and do not wrap the whole output in code fences.